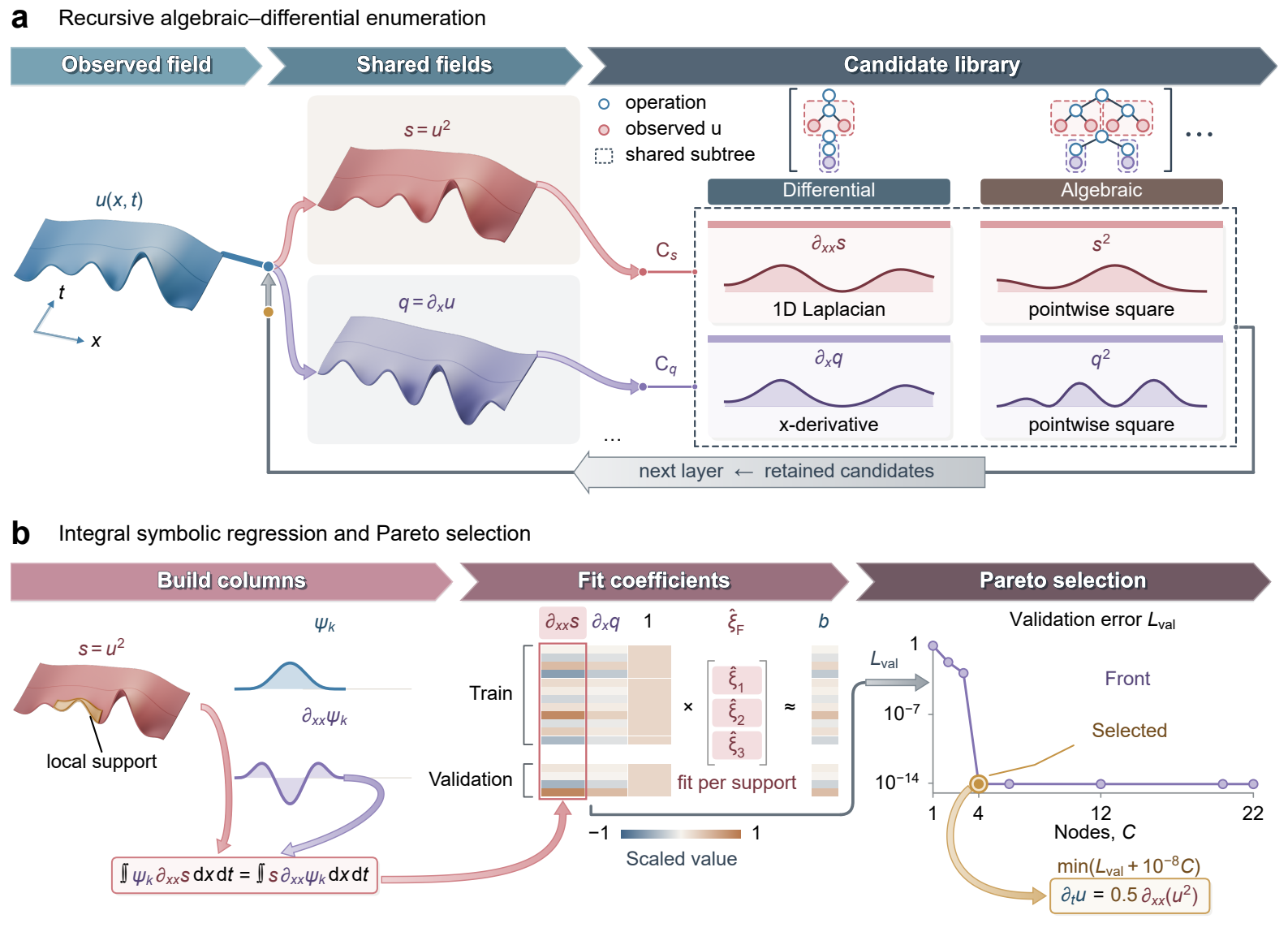

a
Recursive algebraic–differential enumeration
Observed field
Shared fields
Candidate library
operation
observed u
shared subtree
u(x, t)
t
x
s = u²
q = ∂ₓu
Cs
Cq
...
Differential
Algebraic
∂ₓₓs
1D Laplacian
s²
pointwise square
∂ₓq
x-derivative
q²
pointwise square
next layer ← retained candidates
b
Integral symbolic regression and Pareto selection
Build columns
Fit coefficients
Pareto selection
s = u²
local support
ψk
∂ₓₓψk
∫∫ ψk ∂ₓₓs dx dt = ∫∫ s ∂ₓₓψk dx dt
∂ₓₓs
∂ₓq
1
ξ̂F
b
Train
Validation
fit per support
Scaled value
−1
1
Validation error Lval
Lval
Front
Selected
Nodes, C
min(Lval + 10⁻⁸C)
∂t u = 0.5 ∂ₓₓ(u²)